Reading Tea Leaves: Benefits and Costs of Learning about Future Demand Shocks

Finance Theory Group Presentation

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Motivation: Fundamental Shocks vs. Demand Shocks

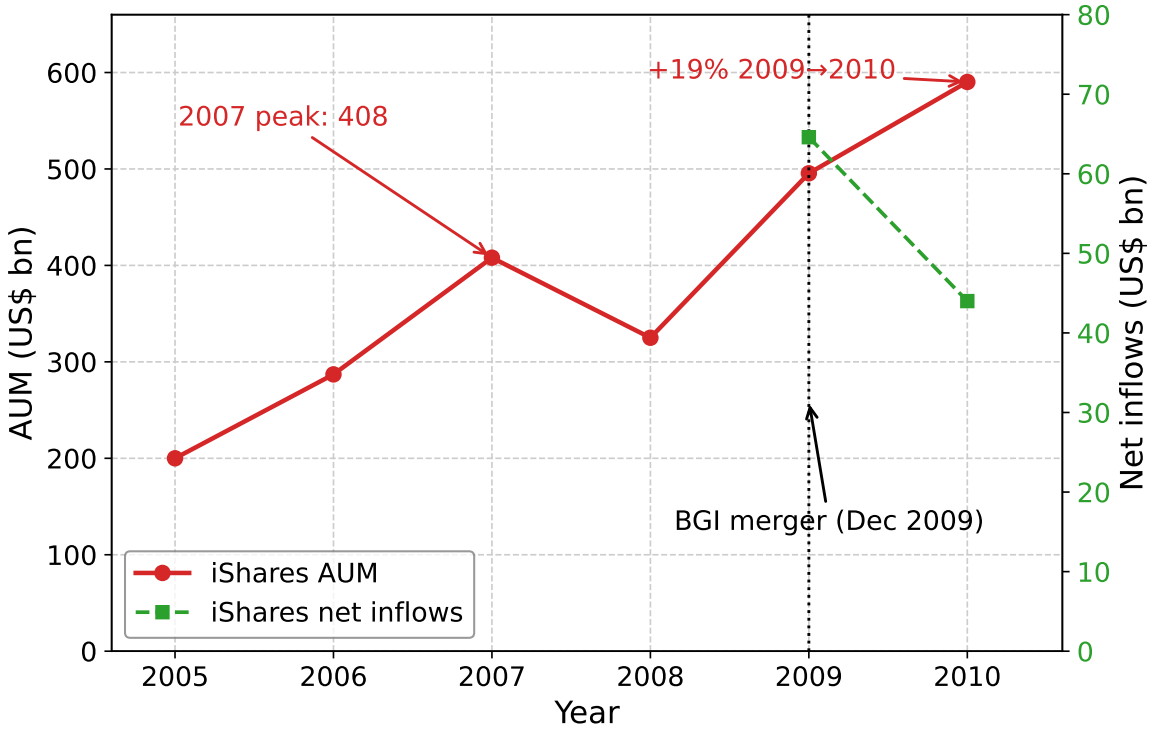
- Fundamental shocks: e.g. dividend innovation
- Demand shocks: orthogonal to fundamentals
  - supply shocks, noise trades
  - hedging needs, liquidity needs, or sentiment

Examples of future demand shocks

- future liquidity crisis
- Han et al. (2016): Index composition change $\Rightarrow$ future demand shock
- Zou (2019): using BlackRock's expansion into the ETF business as an exogenous shock

Example: BlackRock–BGI Transaction

BlackRock announced its acquisition of Barclays Global Investors (BGI), including iShares, on June 11, 2009; the transaction closed on December 1. The event made future iShares expansion foreseeable, creating predictable demand shocks for component stocks that were unrelated to firm fundamentals.



Sources: Barclays PLC Annual Report (2009); BlackRock, Inc. Annual Report (2009); Zou (2019).

Research Question

How does learning about future demand shocks affect investors and securities markets?

- In the static model, learning about future demand shocks can increase price volatility through more heterogeneous private information.
- For a given investor, prices are less informative in the first period but more informative in the second period.
- The future uncertainty channel can improve ex-ante utility
  - Aggregate level: dividend information acquisition $\downarrow \Rightarrow$ dividend risk $\uparrow$ $\Rightarrow$ risk premium $\uparrow$
  - Individual level: lower conditional variance of future excess payoffs

What's Important about FUTURE Demand Shocks?

Fallacy of Composition

- Kurlat & Veldkamp (2015): investors are worse off
  - Individually, more private information is better.
  - On aggregate, asset is less risky with a lower return.

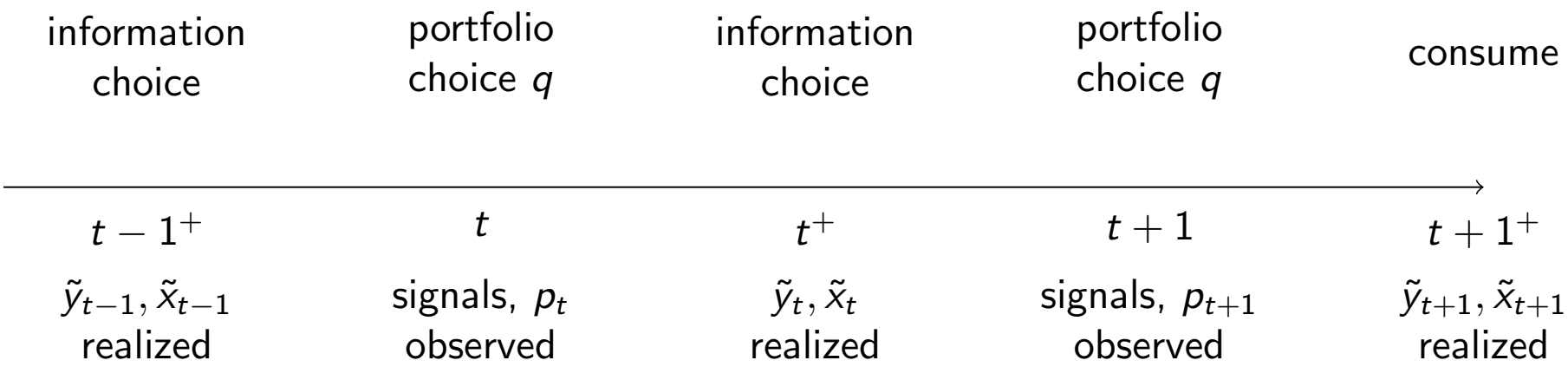
In this paper, learning about future demand shocks creates heterogeneity between generations, breaking the fallacy.

- risk-sharing across time
- information spillover between generations

Literature

	Current	**Future**
Fundamental	Verrecchia (82) Van Nieuwerburgh & Veldkamp (10) Goldstein & Yang (15, 19)	Cai (17)
Demand	Medrano & Vives (04) Ganguli & Yang (09)	This paper

Model: Portfolio Choice with Information Acquisition



Life cycle of Generation-t investors, born at the beginning of period t (period $t-1^+$); $\mathcal{G}$: young, $\hat{\mathcal{G}}$: old.

Model: Portfolio Choice with Information Acquisition

Standard CARA-normal competitive market noisy rational expectations equilibrium (NREE) model with:

Asset:

- A risk-free asset with return r.
- A series of one-period dividend strips $\tilde{d}_t = \mu + g d_{t-1} + \tilde{y}_t$.
- Demand shocks $\tilde{x}_t$ are modeled as shocks to asset supply $\bar{x}_t$ (total supply $\bar{x}_t + \tilde{x}_t$).

Three private signals:

- Signal about current fundamental shock: $\eta_{it}^y = \tilde{y}_t + \varepsilon_{it}^y$.
- Signal about current demand shock: $\eta_{it}^x = \tilde{x}_t + \varepsilon_{it}^x$.
- Signal about future demand shock: $\xi_{it}^x = \tilde{x}_{t+1} + \nu_{it}^x$.

Model: Dividend Strips and Two Channels

- The static model uses a sequence of one-period dividend strips.
- It shuts down the front-running channel and isolates the future uncertainty channel.
- When demand shocks can be fully insured or hedged, it is equivalent to a long-lived-asset model up to constants and multipliers.
- Dividend strips help clarify how information is incorporated into stock prices.

Model: Portfolio Choice with Information Acquisition

OLG investors:

- live for two periods and maximize terminal wealth;
- choose signal precisions $\tau_{\varepsilon yt}$, $\tau_{\varepsilon xt}$ and $\tau_{\nu xt}$ subject to linear information capacity constraint

$$\gamma_y \tau_{\varepsilon yt} + \gamma_x \tau_{\varepsilon xt} + \gamma_{\nu x} \tau_{\nu xt} \leq K$$

  (or a convex constraint);
- make portfolio choices q_{it} and q_{it+1} subject to budget constraint

$$\tilde{W}_{it+1} = r^2 W_0 + r q_{it}(\tilde{d}_t - p_t r) + q_{it+1}(\tilde{d}_{t+1} - p_{t+1} r).$$

Equilibrium price:

- Guess and verify: $p_t = A_t + B_t d_{t-1} + C_t \tilde{y}_t + D_t \tilde{x}_t + E_t \tilde{x}_{t+1}$, with $E_t = 0$ in the static (dividend-strips) case.

Solution Concept

Definition 1 (Equilibrium) A rational expectations equilibrium in this economy is a sequence of information choices $\{\tau_{\varepsilon yt}, \tau_{\varepsilon xt}, \tau_{\nu xt}, \hat{\tau}_{\varepsilon yt+1}, \hat{\tau}_{\varepsilon xt+1}, \hat{\tau}_{\nu xt+1}\}$, prices $\{p_t\}$, and portfolio choices $\{q_{it}, \hat{q}_{it+1}\}$, $\forall t$ such that in each period t,

(a) investors make optimal portfolio choices;

(b) investors make optimal information choices;

(c) asset market clears.

Backward Induction

1. **Step 0:** information choices and portfolio choices can be separated.
2. **Step 1:** At t, the old generation $\hat{\mathcal{G}}$ solve the one-period portfolio choice problem, with private signals from $t-1$ and t.
3. **Step 2:** At $t-1$, the young generation $\mathcal{G}$ solve the two-period portfolio choice problem. This is also the solution for the young generation $\mathcal{G}$ at t.
4. **Step 3:** Market clearing condition $\Rightarrow$ price coefficients.
5. **Step 4:** At t, the old generation $\hat{\mathcal{G}}$ solve the one-period information choice problem, with private signals from $t-1$.
6. **Step 5:** At $t-1$, the young generation $\mathcal{G}$ solve the two-period information problem.

Information Choice: Linear, Equal-Cost Benchmark

Portfolio choice:

$$q_{it} = \frac{\mathbb{E}\left[\tilde{d}_t \mid \mathcal{I}_{it}\right] - p_t r}{\rho r \, \mathbb{V}\mathrm{ar}\left(\tilde{d}_t \mid \mathcal{I}_{it}\right)}$$

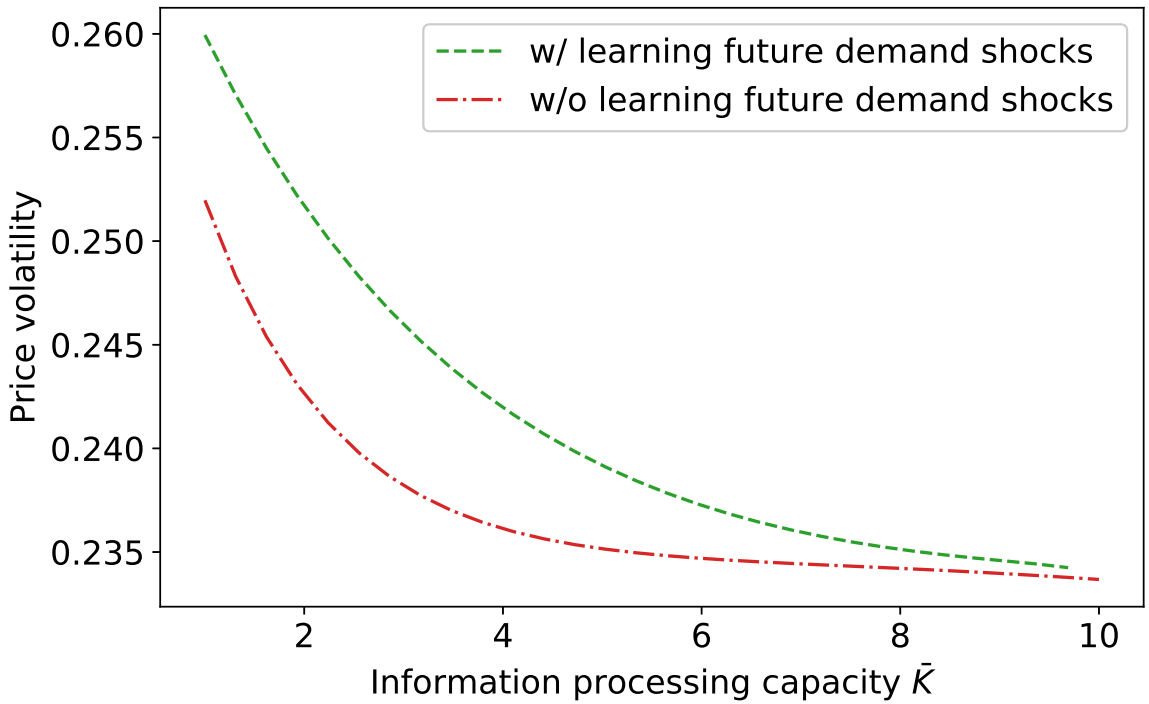
Information choice:

- Under a linear information-capacity constraint with equal signal costs, investors specialize and optimally choose to
  - only learn about future demand shocks in 1st period,
  - only learn about dividend innovation in 2nd period.
- More generally, access to future-demand information shifts precision away from current dividend innovation.

Result 1: Price Volatility (Static Model)

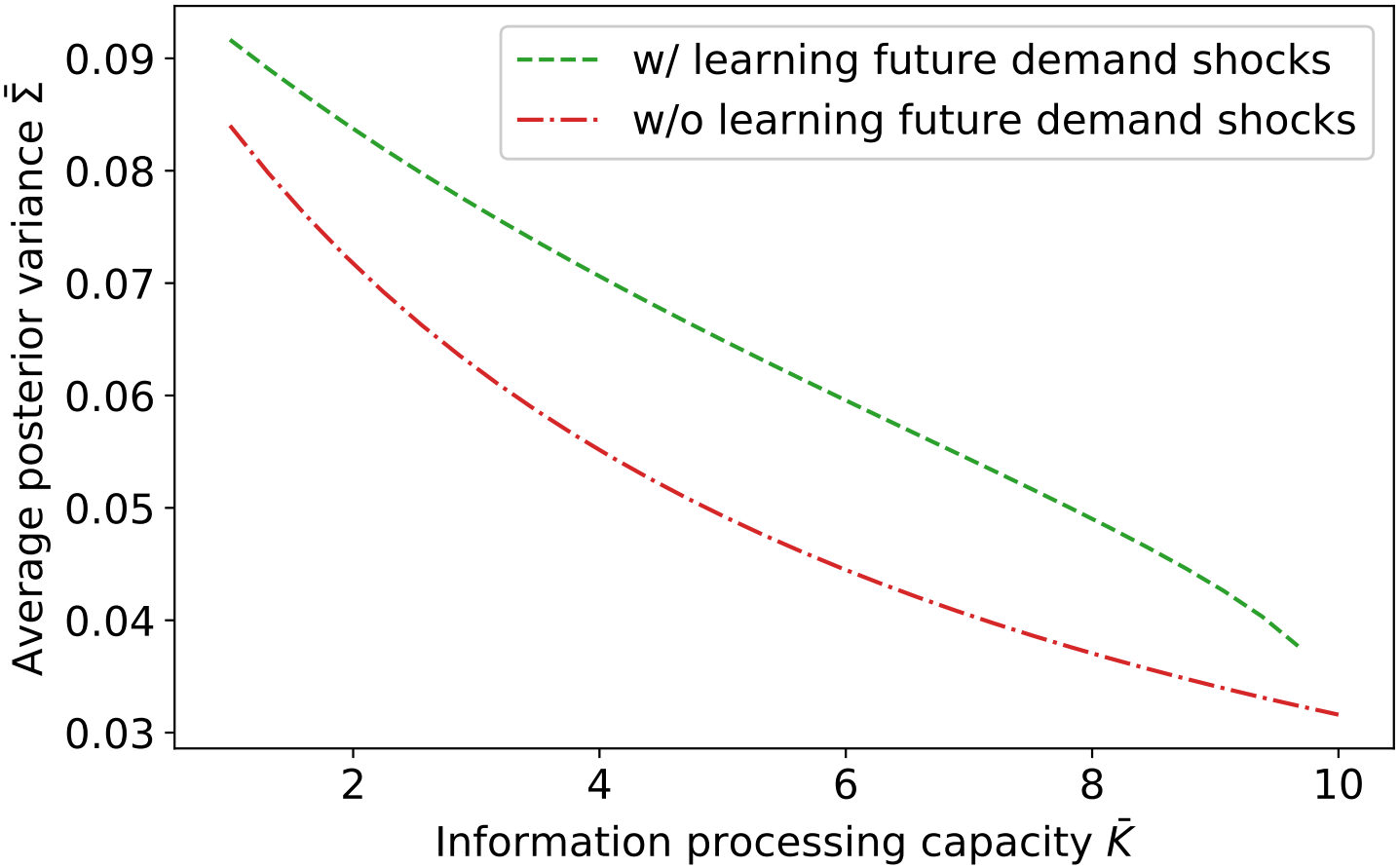
Learning about future demand shocks can increase price volatility $\mathbb{V}\mathrm{ar}(p_t)$ through more heterogeneous private information.

- Extensive margin: two co-existing generations become different
- Intensive margin: the older generation has more private information
- In the baseline calibration, volatility rises when demand-shock precision is low and/or information-processing capacity is high.



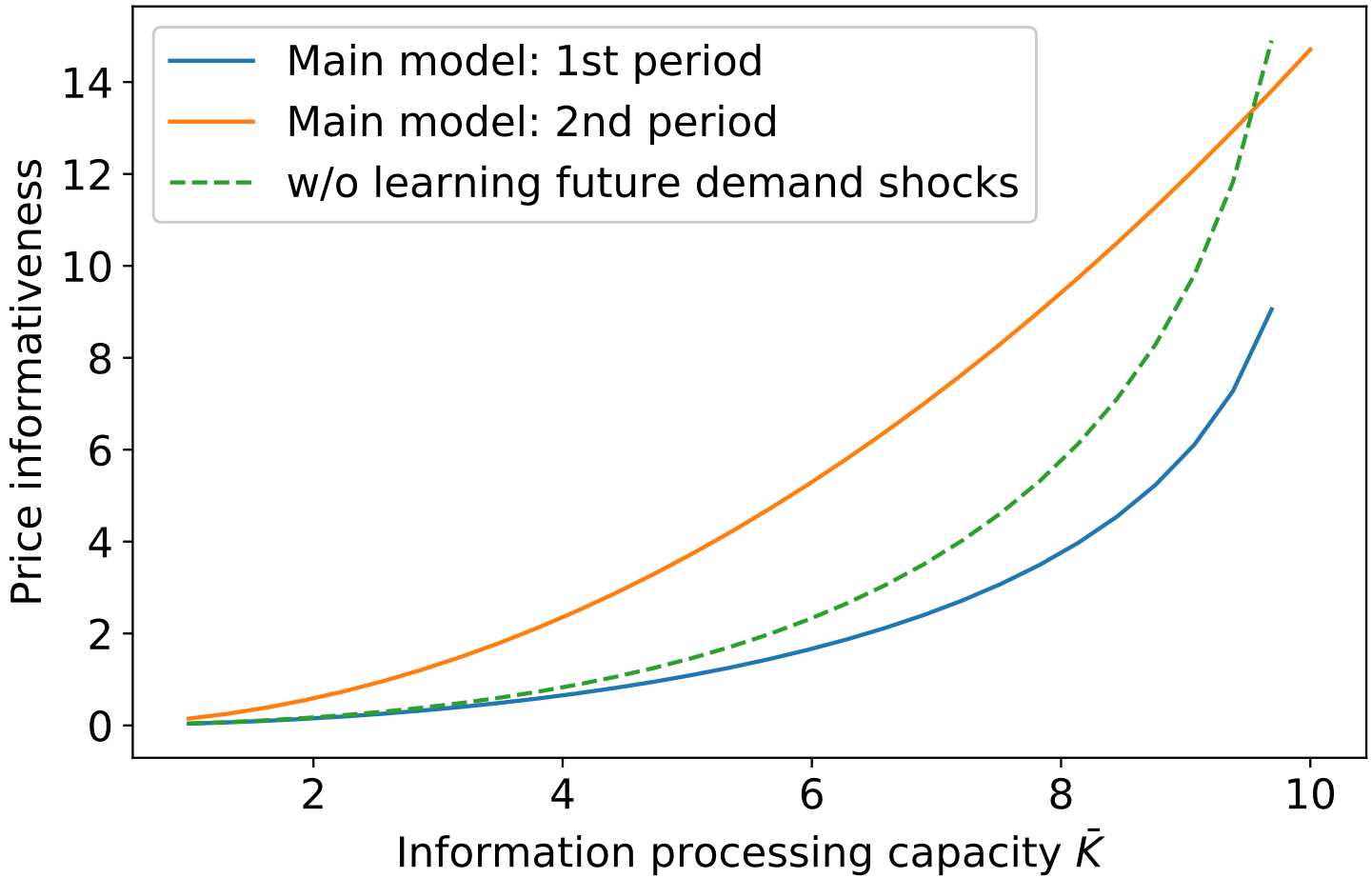
Result 2: Risk Premium (Static Model)

Less fundamental information acquisition $\Rightarrow$ average posterior variance of fundamental shocks, $\bar{\Sigma}$, is larger $\Rightarrow$ higher risk premium.



Result 3: Price Informativeness (Static Model)

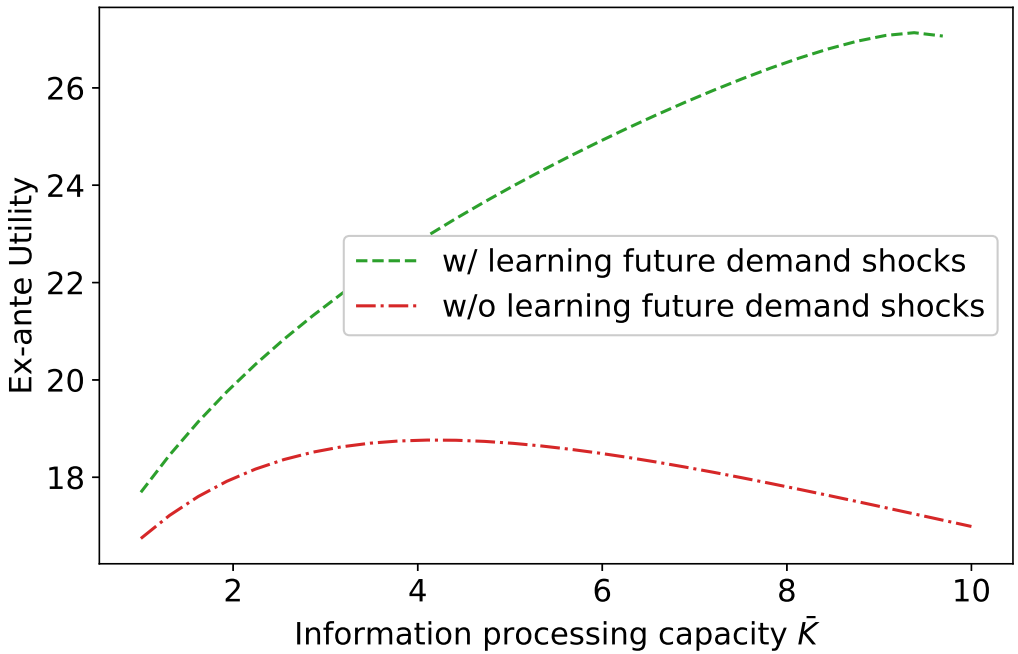
For a given investor, price informativeness decreases in the first period and increases in the second period.



Result 4: Welfare (Static Model)

The future uncertainty channel improves the young generation's ex-ante utility.

- Aggregate level: higher risk premium
- Individual level: take less risk in the second period
- The old generation benefits when information capacity, demand-shock precision, and relative learning costs are sufficiently low.



Concluding Remarks

Benefits:

- The future uncertainty channel raises the young generation's ex-ante utility and can benefit the old generation under suitable parameters.

Costs:

- Price can become more volatile and is less informative contemporaneously.

Dynamic model: The front-running channel operates in the same direction and strengthens these effects.